Problem Set - 12

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1 Relativity

1.1 Photon Rocket

Consider a hypothetical interstellar travel with a photon-propelled spaceship of initial rest mass $M = 1 \times 10^5$ kg. The on-board fuel (antimatter) is annihilated with an equal mass of matter to create photons yielding a reactive force. The matter required for annihilation is collected from the very sparse plasma of the interstellar space (assume that the velocity of the interstellar plasma is zero in the Earth's frame of reference). The speed of light is $c = 3 \times 10^5$ m/s. (NBPhO 2021, Q1)

i) What should be the initial rate μ (kg/s) at which the antimatter should be burned for the acceleration to be equal to the free fall acceleration on Earth?

ii) The engines of the spaceship are switched off when its rest mass has decreased down to $m_f = M/10$, what is its final speed?

iii) The frequency of the emitted photons is measured by an observer on Earth. What is the frequency of the last photons (emitted just before the engine is switched off), as measured on Earth, if the frequency in the the space ship frame remains constant and equal to f_0?

1.2 Annihilation

An electron with kinetic energy 1 MeV travels along the z-axis and collides with a positron at rest. The particles annihilate producing a pair of two photons A and B with equal energies. (NBPhO 20215, Q1)

i) Find the speed of the electron v_e in the lab frame.

ii) Find the energy E_γ of photon A in the lab frame.

iii) Find the momentum p_γ of photon A in the lab frame.

iv) Find the angle α between the z-axis and the momentum of photon A in the lab frame.

v) Why is it not possible for a collision process to produce just one single photon?

The rest mass of electron $m_e = 511$ keV/c^2. You can express your results in electron volts and related units.

2 Optics

2.1 Holographic Lens

Monochromatic light can be focused into a point using an holographic lens (better known as a Fresnel zone plate). This is a thin film which has a system of opaque and transparent concentric rings: the opaque rings obstruct those light waves which would arrive at the focus in the opposite phase (as compared with the light from the transparent rings). In what follows, let us assume that the diameter and focal length of such a lens $d = f = 10$ cm; the wavelength of the monochromatic incident light radiation is $\lambda = 5 \times 10^{-7}$ m.

(NBPhO 2015, Q2)

i) Starting enumeration from the (opaque) centre of the lens, what is the radius of the m-th transparent ring (more precisely, the circle at the middle of the ring)?

ii) Let us consider also a glass lens of the same focal length f and diameter d, such that a parallel incident beam is collected perfectly into the focus (has aspheric surfaces). What is the minimal possible thickness of such a lens if the coefficient of refraction of the glass is $n = 1.5$?

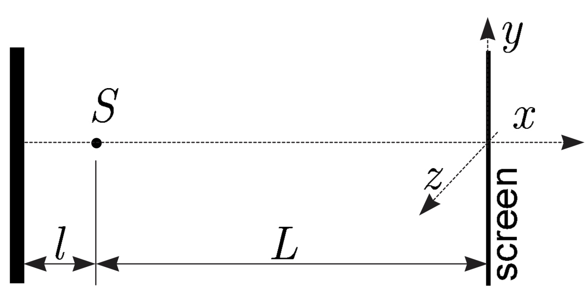
iii) If a short light pulse falls onto a lens, the behavior of the holographic lens is very different from that of the glass lens. Indeed, if we ignore the dispersion of the glass (the dependence of n on the wavelength), the pulse duration at the focus of the glass lens is the same as that of the incident beam. Meanwhile, in the case of a short pulse ($\tau = 3 \times 10^{-14}$ s), the pulse duration at the focus of the holographic lens becomes significantly dilated. Sketch qualitatively the light intensity I at the focus of the holographic lens as a function of time. A scale is required for time t, but not for I. The speed of light $c = 3 \times 10^8$ m/s.

iv) Even if the light source is an ideal laser providing a perfectly monochromatic radiation, the short pulse is no longer strictly monochromatic. Estimate the width of the range of wavelengths in the pulse of duration τ.

v) Estimate the duration of this light pulse in the focus of the glass lens. Assume that in the glass, the light wave group velocity v_g depends on the wavelength at the rate $\mathrm{d}v_g/\mathrm{d}\lambda = 0.02 \times v_g/\lambda$ and that for the wavelength λ, group speed is equal to the phase speed.

2.2 Mirror Interference

A point source S emits coherent light of wavelength λ isotropically in all directions; thus, the wavefronts are concentric spheres. The waves reflect from a dielectric surface placed at a distance $l = N\lambda$ (where N is a large integer) from the point source, and the interference pattern is observed on a screen which is placed to a distance $L \gg l$ from the point source (see figure).



In what follows we use the x, y, and z coordinates as defined in the figure. The screen is parallel to the mirror and lies in the $y - z$-plane.

i) At which values of the y-coordinate (for $z = 0$) are the interference maxima observed on the screen? You may assume that $y \ll L$.

ii) Sketch the shape of few smallest sized interference maxima on the screen (in $y - z$-plane)

iii) Now the flat screen is replaced with a spherical screen of radius L, centered around the point source. How many interference maxima can be observed?

3 Modern Physics

3.1 Spin System

Let us consider a system of N independent magnetic dipoles (spins) in a magnetic field B and temperature T. Our goal is to determine some properties of this system by using statistical physics. It is known that the energy of a single spin is $E = \epsilon m$, where $m = \pm 1/2$ and $\epsilon = \alpha B$. (NBPhO 2014, Q7)

i) What is the probability $p_\uparrow$ for a spin to be in exited state, i.e. have positive energy?

ii) What is the average value of the total energy E_s of the spin system as a function of B and T?

iii) Using high temperature approximation $T \gg \alpha B m/k$, simplify the expression of E_s.

iv) Using high temperature approximation $T \gg \alpha B m/k$, find the heat capacity C of the spin system.

Q2. Sources, atoms, and spectra

A. Light source

In its proper reference frame, a point source emits light in the form of a divergent conical beam, with the angular width of 90° (from -45° to +45° with respect to the cone axis). In a reference frame which moves towards the source with an unknown speed v, the angular width of the beam is of only 60° (from -30° to +30° with respect to the same cone axis). The light speed in vacuum is $c = 2.998 \cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}}$.

A	Determine the speed v of the source.	2.50 p.

B. Balmer emission spectrum

The spectral resolving power of a spectrometer is $R = 5 \cdot 10^5$. The spectrometer is used to observe the Balmer series in the emission spectrum of the hydrogen atom (the visible domain).

Note: The possible mechanisms of broadening of the spectral lines (Lorentzian, Gaussian, etc.) will not be considered.

B.1	Express the mathematical definition of the spectral resolving power of the instrument.	0.25 p.

B.2	Determine the highest value for the principal quantum number n of the energy level for which the spectral line emitted by an atom for the transition to the level $n' = 2$ can still be distinctly resolved by the instrument, with respect with its neighbours.	2.25 p.

C. Absorption spectra

The energy levels of an atom are given by $E_n = -\frac{A}{n^2}$, where n is an integer and A is a positive constant. Among the adjacent spectral lines which, at room temperature, the atom can absorb, two have the wavelengths 97.5 nm and 102.8 nm, respectively. The elementary electric charge is $e = 1.602 \cdot 10^{-19}$ C, the speed of light in vacuum is $c = 2.998 \cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}}$, and Planck's constant is $h = 6.626 \cdot 10^{-34}$ J·s.

C.1	Find the values of the quantum numbers n of the energy levels implied in the transitions.	3.00 p.

C.2	Determine the value of the constant A in joule and in electron-volt.	1.50 p.

C.3	Identify the nature of the atom and justify the choice made.	0.50 p.